PLUS 9000 HB

One Pot Photo-polymer Textile Emulsion

Ready-to-use photopolymer emulsion for textile printing

DESCRIPTION

Plus 9000 HB is a high-quality, ready-to-use screen-printing emulsion based on pure SBQ photopolymer, red in colour. The emulsion has a very high solids content (47%), a very short exposure time, and is intended for use with non-water-based textile inks.

FEATURES AND ADVANTAGES

Properties	Advantages
One-component system	Safer application; no sensitisation required.
Excellent coating quality	Easy movement of the coating trough over the emulsion provides fast, uniform coating with minimal effort.
Very short exposure time	Significant time savings when producing large volumes of screens.
High solids content, 47%	A smaller number of emulsion coats provides the required coating thickness, even on coarse mesh. Compared with other SBQ emulsions, Plus 9000 provides a 25–50% higher stencil profile with the same number of coats, depending on mesh.
Wide exposure latitude; very high resolution (up to 100 µm) and sharp image edges	Up to 25% overexposure is permitted, which is significant for fast emulsions. Even the optimum exposure time provides very high resolution. No special procedures are required to increase resolving power. Edge sharpness is subjective, but tests cited in the source state that Plus 9000 significantly outperforms competing products on this measure.
Easy reclaiming	Unlike many SBQ-based emulsions, Plus 9000 is easily removed from screens. It is not prone to hardening in the mesh or “locking” under the influence of solvents. (The source advises avoiding aggressive solvents.)

APPLICATION

Plus 9000 is designed for printing textile products when using the following ink types:

- Non-water-based plastisols
- Puff and other similar non-water-based inks

INSTRUCTIONS FOR USE AND SAFE HANDLING

Unlike other photosensitive emulsions, PLUS 9000 HB does not require sensitisation before use.

MESH PREPARATION

New polyester mesh: Alkemi Autoprep Gel and Alkemi Universal Mesh Prep, or Alkemi Degreaser Concentrate.
Used polyester mesh: Alkemi Universal Mesh Prep or Alkemi Degreaser Concentrate.

COATING AND DRYING

Position the screen on its edge, angled away from the operator, and use a good-quality coating trough:

1. Apply one coat to the outside (printing side) of the screen.
2. Apply one coat to the inside (squeegee side) of the screen.
3. Dry the screen horizontally, with the squeegee side facing upwards, and store in darkness or under safe yellow light. Dry at a temperature not exceeding 30°C.
4. Additional coats may be applied to the underside (printing side), with intermediate drying.

EXPOSURE

Correct exposure is essential for an optimum stencil. Underexposure will produce a weak stencil and reduce its durability. Remove dust from the stencil surface and secure the positive film with adhesive tape. Place the entire screen in a suitable vacuum exposure frame and expose. The source recommends using the Autotype Exposure Calculator to measure lamp output and calculate the correct exposure time.

EXPOSURE TIME (*IN MINUTES)

Lamp	Distance	0–10 minute scale in original chart
2 kW metal-halide lamp	120 cm	Blue cells: 4, 5, 6
3 kW metal-halide lamp	120 cm	Blue cells: 3, 4
5 kW metal-halide lamp	120 cm	Blue cells: 2, 3
125 W mercury lamp	50 cm	Blue cells: 7, 8, 9

Lamp	Distance	0–10 minute scale in original chart
UV fluorescent lamp	10 cm	Blue cells: 6, 7, 8
Thorn Graph X	120 cm	Blue cells: 5, 6, 7

Based on results for white polyester monofilament mesh, 43 threads/cm (110 threads/inch), with a 2+2 coating.

WASHOUT AND DRYING

Wash the screen on both sides with a strong stream of water. Continue rinsing until all image areas are clean and clearly open. Remove excess moisture with a non-woven material and dry at a temperature not exceeding 30°C (86°F). Cover pinholes or defects with sensitised emulsion (for water-based inks) or Alkemi Blue Block (for UV or solvent-based inks).

STENCIL REMOVAL

Remove ink from the screen using Alkemi Screen Cleaner. Apply Stencil Remover Concentrate to both sides of the screen with an Alkemi Brush and leave for 2–4 minutes. The stencil can then be removed with a strong stream of water. If stains or ghost images remain, use Alkemi Haze Remover HV or Alkemi Autohaze.

FIRST AID

Eye contact: Rinse with water for 10 minutes.
Skin contact: Wash thoroughly.
If swallowed: Dilute with water and seek medical attention.

DISPOSAL

Small quantities may be washed into domestic drainage. Larger quantities must be disposed of through an authorised industrial-waste disposal company.

PACKAGING

PLUS 9000 HB is supplied in sealed plastic containers in the following packs: 6 × 1 litre, 4 × 4 litres and 1 × 200 litres.

STORAGE CONDITIONS AND SHELF LIFE

Store the emulsion at 3–20°C. Shelf life under these conditions is 24 months. Shelf life will be reduced at higher temperatures.

SAFETY AND PRECAUTIONS

MacDermid Autotype recommends that the company/operator read the product Safety Data Sheet and follow the relevant safe-use precautions before using the product.

WASTE MANAGEMENT

Before applying any waste-management recommendations from MacDermid Autotype, the user must review the applicable local, national and federal requirements for on-site processing or off-site disposal, which may require permits. Where any discrepancy exists, local, national and federal regulations take precedence.

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